RA Trial Task Instructions (Week 1)

MAR Missing-Data Simulation · Listwise Deletion · Regression

Overview

Each RA completes **10 papers**. The workflow for every paper is:

`Prepare data` → `Introduce MAR missingness` → `Listwise Deletion` → `Run regression` → `Record results`

The only experimental variable is the **missing-data proportion**. Everything else is held constant: one missingness mechanism (MAR), one handling method (Listwise Deletion), and one run per combination (no repeated draws).

Step-by-Step Procedure

Step 1: Understand the Regression Specification

Open the replication `.do` file and locate the **main regression**—typically the most complete column in the main table (i.e. the column that includes all controls and fixed effects). Record the following:

- Dependent variable (Y)
- Key independent variable(s) (key X)
- Control variables
- Regression type (OLS / FE / IV, etc.)
- Fixed effects and clustering of standard errors
- Sample restrictions

Step 2: Data Preparation

Replicate every variable transformation in the `.do` file (logs, interactions, winsorising, etc.) and apply the original sample restrictions to obtain an **analysis sample identical to the one used in the paper**.

Save this as `baseline_data.csv`.

Step 3: Run the Baseline Regression

Using `baseline_data.csv`, run the paper's main regression exactly as specified. Confirm that your results match the published results (coefficient sign, significance level, order of magnitude). Record this as the **baseline result**.

Stop and check if your baseline results diverge noticeably from the paper. Do not proceed until the discrepancy is resolved.

Step 4: Select Key Variables (3–5)

Choose **3–5 variables** from the regression as targets for introducing missingness.

Select	Do NOT select
Continuous numerical variables	Dependent variable
Key independent variable(s)	ID / time identifiers
Important control variables	Binary dummy variables

Step 5: Select the MAR Auxiliary Variable (1)

Choose one variable that satisfies all of the following: (i) numerical, (ii) has little or no missingness, and (iii) is correlated with most of the key variables selected in Step 4.

The auxiliary variable **must not** be one of the key variables.

Its role: observations with larger values of the auxiliary variable will have a higher probability of being set to missing.

Step 6: Introduce MAR Missingness

All RAs use the following parameters:

Parameter	Value
Missingness mechanism	MAR only
Missing proportions	1%, 5%, 10%, 20%, 30%, 40%, 50% (7 levels)
MAR strength	1.5
Random seed	`seed = 2026` (uniform across all RAs)

How MAR works. For each key variable, every observation's probability of being set to missing is proportional to the auxiliary variable raised to the power of 1.5. The probabilities are then rescaled so that the total fraction of deleted values equals the target proportion.

Each key variable $\times$ each missing proportion $\rightarrow$ produces **1 dataset with missingness**.

Step 7: Apply Listwise Deletion

For each dataset with missingness, **drop all rows where the target key variable is missing**. Save the resulting dataset.

Step 8: Run Regression and Record Results

On each post-deletion dataset, run **exactly the same regression as in Step 3**—same dependent variable, same independent variables, same controls, same fixed effects, same clustered standard errors.

Record the following for every regression in `regression_results.xlsx`:

Field	Description
Key variable	Name of the variable with induced missingness
Missing proportion	1% / 5% / 10% / 20% / 30% / 40% / 50%
Post-deletion N	Number of observations after listwise deletion
Coefficient ($\hat{\beta}$)	Point estimate of the key X
Standard error (SE)	Corresponding standard error
t-value / z-value	Test statistic
p-value	Corresponding p-value
Significance flag	`***` ($p < 0.01$), `**` ($p < 0.05$), `*` ($p < 0.1$), `ns`
R^2	Regression R^2 (if applicable)
Consistent with baseline?	Whether sign and significance match baseline (Yes / No)

Each paper produces $K \times 7$ result rows (where K is the number of key variables), plus 1 row for the baseline.

Dataset and Result Counts

Let K denote the number of key variables.

Stage	Formula	K=4 example
Datasets with missingness	$K \times 7$	28
Post-LD datasets	$K \times 7$	28
Regression results (rows)	$K \times 7 + 1$ (incl. baseline)	29
Total files per paper	$2K \times 7 + 2$	**58**

For 10 papers with $K = 4$: approximately 580 data files + 10 result files.

Deliverables

Folder Structure

```
Paper_AuthorYear/
  paper_info.xlsx              # Paper metadata + variable choices
  regression_results.xlsx      # All regression results
  baseline_data.csv            # Complete analysis sample
  missing/                     # Datasets with MAR missingness
    var1_MAR_01pct.csv
    var1_MAR_05pct.csv
    var1_MAR_10pct.csv
    var1_MAR_20pct.csv
    var1_MAR_30pct.csv
    var1_MAR_40pct.csv
    var1_MAR_50pct.csv
    var2_MAR_01pct.csv
    ...
    var4_MAR_50pct.csv
  listwise/                    # After Listwise Deletion
    var1_MAR_01pct_LD.csv
    var1_MAR_05pct_LD.csv
    ...
    var4_MAR_50pct_LD.csv
```

File naming convention:

- With missingness: `{varname}_MAR_{proportion}.csv`
- After deletion: `{varname}_MAR_{proportion}_LD.csv`

`paper_info.xlsx` — **Required Fields**

Field	Content
Paper title	Full title
Dependent variable	Variable name + description
Key independent variable(s)	Variable name(s) + description
Control variables	Complete list
Regression type	OLS / FE / IV, etc.
Fixed effects	Individual, time, industry, etc.
Clustered SEs	Clustering level
Sample restrictions	Specific filtering logic
Key variables (3–5)	Target variables for missingness
MAR auxiliary variable	Variable name + description
Complete sample size N	Row count of `baseline_data.csv`
Published coefficient	Key X coefficient reported in the paper
Baseline coefficient	Key X coefficient from your replication

`regression_results.xlsx` — **Format**

One Excel file per paper. Each row is one regression result. The first row is the baseline; subsequent rows correspond to each (key variable × missing proportion) combination.

Column order: Key Variable, Missing Proportion, Post-LD N, $\hat{\beta}$, SE, t-value, p-value, Significance Flag, R^2, Consistent with Baseline.

Pre-Submission Checklist

Baseline regression results approximately match the published paper (sign, significance, magnitude).
3–5 key variables selected, all continuous.
MAR auxiliary variable is **not** among the key variables.
All seeds set to `2026`.
`missing/` folder: 7 files per key variable (one per proportion).
`listwise/` folder: 7 files per key variable (one per proportion).
Every LD file has fewer rows than `baseline_data.csv`.
`regression_results.xlsx` has $K \times 7 + 1$ rows (including baseline).
All regressions use the **identical** specification as the baseline.
`paper_info.xlsx` is fully completed.
`baseline_data.csv` is the complete sample (no artificial missingness).

FAQ

Q: Which regression should I use?
A: The most complete column in the main table—usually the one with all controls and fixed effects included.

Q: I can't understand or run the `.do` file.
A: Document the specific problem and contact me. Do not skip ahead.

Q: My baseline results don't match the paper exactly.
A: Minor differences in decimal places are acceptable. If the coefficient sign differs or significance changes substantially, your data processing likely has an error. Stop and check, or contact me.

Q: After 50% deletion the sample is very small and the regression won't converge.
A: In `regression_results.xlsx`, note "regression did not converge" or "insufficient sample size" in the corresponding row. Leave the coefficient and related fields blank.

Q: Does Listwise Deletion drop rows where *any* variable is missing, or only the target variable?
A: Drop only the rows where the **target key variable** is missing. Each LD file corresponds to missingness in one specific variable.

Q: What if the original data already contains missing values?
A: Start from complete observations—rows where all regression variables are non-missing. That is your `baseline_data.csv`. Introduce artificial missingness on top of that.

Q: Which software should I use for the regressions?
A: Stata or Python are both fine; stay consistent with the original paper's setup. As long as the specification is identical, results will not differ materially across software.

---

10 papers per person. Due Friday. Questions welcome anytime.